

Day 3 Handout: Observational Data and Quasi-Experiments

Experimentation and Causal Inference, SDAIA Academy. Revision notes for participants.

What Day 3 was about

An observational study is an experiment you did not run. Without randomization, treated and control units differ for reasons other than treatment, and every method on Day 3 is an attempt to recover the balance that randomization would have given for free. Each pays for it with an assumption; your job is to name that assumption, argue for it, and probe it.

Confounding and DAGs

A directed acyclic graph (DAG) draws your assumptions about the data-generating process. Nodes are variables; an arrow $X \rightarrow Y$ says X can directly cause Y ; a missing arrow is also a claim. Association flows along paths. Causal paths ($T \rightarrow \dots \rightarrow Y$) are what we want; backdoor paths (starting with an arrow into T) are what we must block.

Three elementary structures:

Structure	Shape	Behavior	Rule
Chain (mediator)	$T \rightarrow M \rightarrow Y$	M transmits the effect; conditioning on M blocks part of it	Do not adjust when you want the total effect
Fork (confounder)	$T \leftarrow X \rightarrow Y$	X opens a backdoor; conditioning on X blocks it	Adjust
Collider	$T \rightarrow C \leftarrow Y$	Blocked by default; conditioning on C opens a spurious path	Do not adjust

Backdoor criterion: choose a set Z that blocks every backdoor path from T to Y and contains no descendant of T . Adjusting for Z identifies the effect. This is the rule that decides what goes into a regression or a propensity model; fit does not.

Collider bias in the wild: selection is conditioning. Studying only hospitalized patients, only funded startups, or only completed Injaz sessions conditions on a collider and can manufacture association from nothing. “Control for everything” is dangerous advice.

Procedure: draw the DAG from domain knowledge before looking at correlations; list backdoor paths; find a blocking set; exclude mediators, colliders and post-treatment variables; confirm every chosen covariate is measured; record the graph with the analysis.

Injaz training DAG: age, digital literacy, prior completions and region cause both enrolment and completion (confounders, adjust). Enrolment raises support calls, which raise completion (mediator, exclude). The lab showed the correctly adjusted effect at about +4 points and the mediator-controlled “effect” at about +2, which answers a different question (the direct effect not via support).

Adjustment estimators

All target the same estimand under ignorability given X ; they differ in what they model and how they fail.

Outcome regression (g-formula): model $E[Y \text{ given } T, X]$; predict both potential outcomes for every unit; average the difference. Fails if the outcome model is misspecified.

Matching: pair treated and control units with similar X (or similar propensity, within a caliper such as 0.2 times the SD of the logit); average within-pair differences for the ATT. Transparent; discards unmatched units (which changes the estimand); sensitive to the distance metric and overlap.

Propensity score $e(x) = P(T = 1 \text{ given } X)$. Rosenbaum and Rubin: if adjusting for X suffices, adjusting for the scalar $e(x)$ suffices. Judge the propensity model by the covariate balance it achieves (standardized mean differences

below 0.1), not by its AUC. A perfect classifier destroys overlap.

Inverse-probability weighting (IPW): treated units get weight $1/e(x)$, controls $1/(1 - e(x))$, rebuilding a pseudo-population where T is independent of X . Propensities near 0 or 1 create enormous weights: $e = 0.05$ gives weight 20. Stabilize (multiply by the marginal treatment probability), trim to common support, or use overlap weights. Always print the maximum and 99th-percentile weight.

Doubly robust (AIPW, TMLE): combine an outcome model and a propensity model; consistent if either is correctly specified. With cross-fitting (fit nuisance models on one fold, predict on the other) you can use flexible machine learning and keep valid intervals. The modern default.

Observational checklist, in order: define the estimand and target population; adjustment set from the DAG; check overlap and trim; verify balance after adjustment; prefer a doubly robust estimator with cross-fitting and report an interval; follow with a sensitivity analysis.

Difference-in-differences

When a treatment switches on at a known time for some units, compare their before/after change to the change in untreated units:

$DiD = (Y_{\text{treated,post}} - Y_{\text{treated,pre}}) - (Y_{\text{control,post}} - Y_{\text{control,pre}})$

Worked 2 by 2: treated 40 to 52 (change 12), control 38 to 44 (change 6), $DiD = +6$. The regression form has unit and time fixed effects plus a treated-times-post interaction (two-way fixed effects, TWFE); cluster standard errors at the unit of treatment.

Parallel trends: absent treatment, the treated group would have moved in parallel with the control group. It is about a counterfactual, so it is never fully testable. Support it with an event study: estimate a coefficient for each period relative to adoption (omit $t = \text{minus } 1$); flat pre-period coefficients then a post jump is the signature of a credible effect. A shock that coincides with treatment still breaks it.

Staggered adoption: naive TWFE uses already-treated units as controls for later adopters and can be badly biased, even wrong-signed, under heterogeneous effects. Use never-treated or not-yet-treated comparisons, or heterogeneity-robust estimators (Callaway-Sant'Anna, Sun-Abraham, de Chaisemartin-D'Haultfoeuille).

Placebo tests: pretend treatment happened earlier, using pre-period data only; a significant "effect" falsifies parallel trends.

Synthetic control: one treated unit and many donors; choose non-negative weights so the synthetic unit matches the treated unit's pre-treatment path; the post-period gap is the effect; inference by placebo on untreated units.

Injaz reminders: a naive before/after in treated regions showed about +7 points because national adoption was rising anyway; the DiD gave about +3 with flat pre-trends and a null placebo.

Instrumental variables

An instrument Z rescues identification when treatment is confounded, if three conditions hold: relevance (Z shifts T ; testable via the first-stage F-statistic, with F below about 10 signaling a weak instrument), exclusion (Z affects Y only through T ; untestable), independence (Z is as good as random with respect to the confounders).

Two-stage least squares: regress T on Z , then Y on predicted T . The Wald ratio is the simplest form: effect of Z on Y divided by effect of Z on T . Worked example: reduced form 0.9 points, first stage 0.30, $LATE = 3.0$.

What IV identifies: the local average treatment effect (LATE), the effect for compliers, the units whose treatment the instrument moves. It says nothing about always-takers or never-takers, and a different instrument gives a different, equally valid LATE. LATE is not the ATE.

Injaz instrument: distance to the nearest in-person service center for online-versus-in-person filing. Relevance is strong; exclusion must be argued, not assumed.

Regression discontinuity

When treatment is assigned by a running variable crossing a cutoff, units just above and just below are comparable: a local experiment. Sharp RDD: crossing the cutoff determines treatment. Fuzzy RDD: it changes the

probability (IV at the boundary). The estimate is the jump in fitted outcomes at the cutoff (worked example: 44 versus 40, local effect +4). Validity checks: no bunching of the running variable at the threshold (McCrary density test), pre-treatment covariates continuous through the cutoff, stability across bandwidths and polynomial orders. The estimand is local to the cutoff.

Choosing a design

Is there a threshold rule that assigns treatment? Consider RDD. Is there a variable that shifts uptake but not the outcome directly? Consider IV. Did treatment switch on at a known time for some units? Consider DiD or synthetic control. Is unconfoundedness plausible given rich covariates? Consider doubly robust adjustment. Always ask which assumption is least implausible here and how you would probe it.

Lab 3 recap

Naive gap about +12 points. g-formula about +4.0; stabilized IPW about +4.5 with well-behaved weights after trimming to common support; cross-fit AIPW about +4.5 with CI roughly [3.0, 6.0]. The planted truth was about +4.5. The mediator-controlled model gave +2.2: the wrong answer to the question asked. DiD on the reminder rollout: about +3.0 with CI [2.4, 3.6], flat pre-trends, placebo about 0.

Discussion scenario cards (Hour 5)

Card A. A ministry gives a training subsidy to firms with more than 50 employees, starting in 2024. Firm-level employment data 2020 to 2026 exist for all firms.

Card B. Citizens closer to a service center are more likely to file in person. You want the effect of filing channel on completion.

Card C. A reminder program was switched on in five regions in March and the other eight in September. Monthly regional completion rates exist.

Card D. A voluntary workshop with rich pre-enrolment covariates (age, literacy, prior usage, region). No timing variation, no threshold.

For each: name the design, the identifying assumption, and one diagnostic you would run.

Self-check (answers at the end)

1. In T from X to Y , what is X ?
2. What must a valid backdoor adjustment set never contain?
3. A doubly robust estimator is consistent if what holds?
4. Propensity scores near 0 or 1 in IPW cause what?
5. State the DiD identifying assumption and how you support it.
6. Why can classic TWFE mislead under staggered adoption?
7. Name the three IV conditions and what IV identifies.
8. A spike in the running-variable density at the cutoff suggests what?

Answers

1. A confounder.
2. A descendant of T (a mediator or collider).
3. Either the outcome model or the propensity model is correctly specified.
4. Extreme weights and high variance.
5. Parallel trends; supported (not proven) by near-zero pre-period event-study coefficients and null placebo tests.
6. It uses already-treated units as controls for later adopters.
7. Relevance, exclusion, independence; the LATE for compliers.
8. Manipulation or sorting at the threshold.

Scenario cards: A is RDD at the 50-employee cutoff (check bunching and covariate continuity) and also DiD from 2024 with never-treated firms (check pre-trends); B is IV with distance as the instrument (report first-stage F , argue exclusion); C is DiD with the September regions as not-yet-treated controls (event study, placebo timing,

staggered-robust estimator); D is doubly robust adjustment on the DAG-derived backdoor set (overlap, balance, sensitivity analysis).